

OLS and Regression Assumptions

POLS 602
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Review

1. Summarize how the OLS estimator works
2. What do the normal equations say about the residuals and the columns of $\mathbf{X}$?

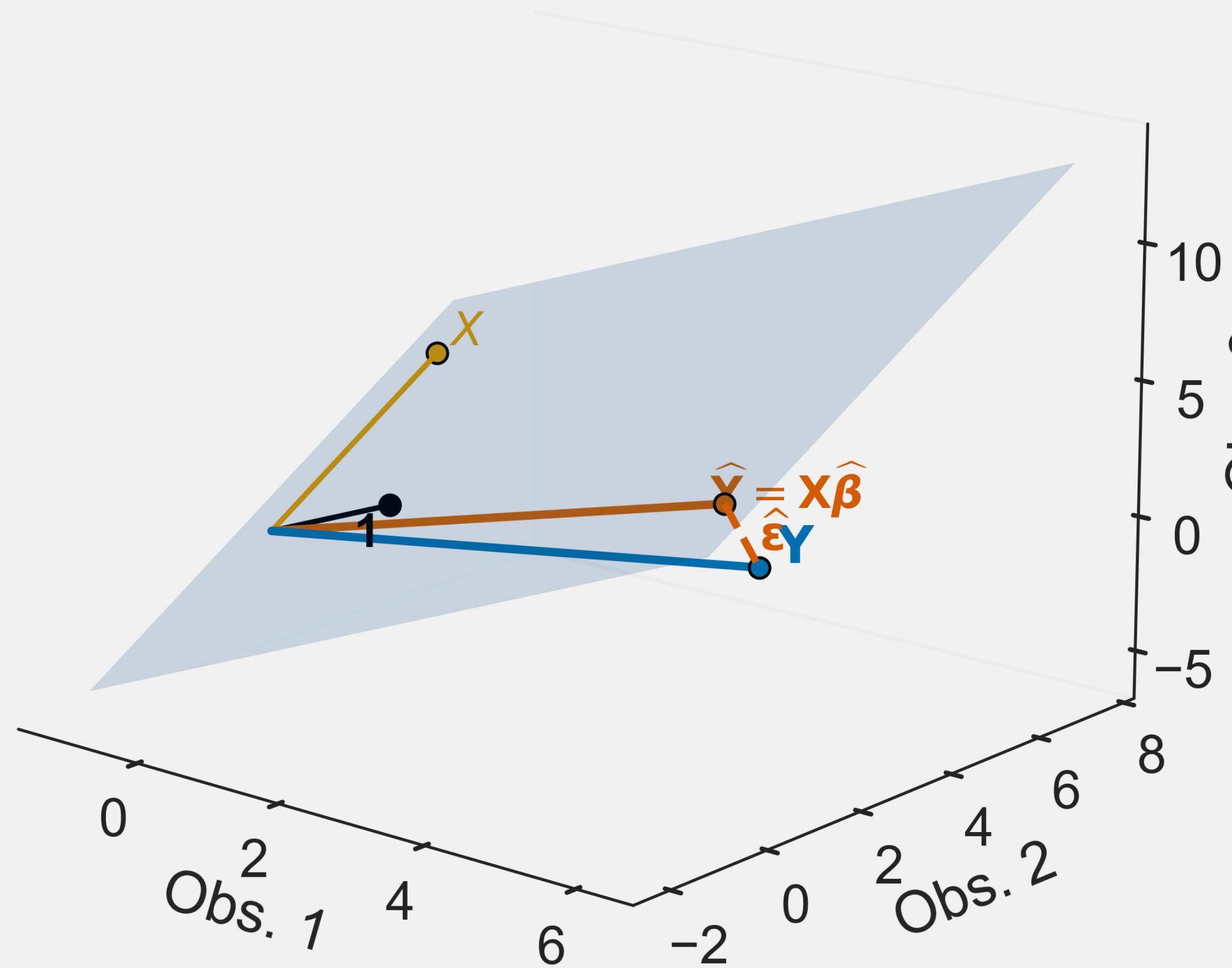
$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}$$

3. What is the matrix formula for the OLS estimator?

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

Regression as Projection

Column Space of X (the shaded plane), Y , and $\hat{\epsilon} \perp$ the Plane



Today

- Why is OLS the best linear unbiased estimator?
- What assumptions have to hold for that to be true?
- What additional assumptions do we need for statistical inference?

OLS Is BLUE

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Under a set of assumptions, OLS is the **B**est **L**inear **U**nbiased **E**stimator of β .

Three properties, each of which describes the estimator $\hat{\beta}$:

- **Best**: smallest variance in the sampling distribution of $\hat{\beta}$
- **Linear**: linear in its parameters
- **Unbiased**: correct on average

Linear vs. Nonlinear Equations

The model is linear in its parameters.

Linear in the parameters:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i$$

$$Y_i = \beta_0 + \beta_1 \log(X_i) + \varepsilon_i$$

Not linear in the parameters:

$$Y_i = \beta_0 + \beta_1^2 X_i + \varepsilon_i$$

$$Y_i = \beta_0 + X_i^{\beta_1} + \varepsilon_i$$

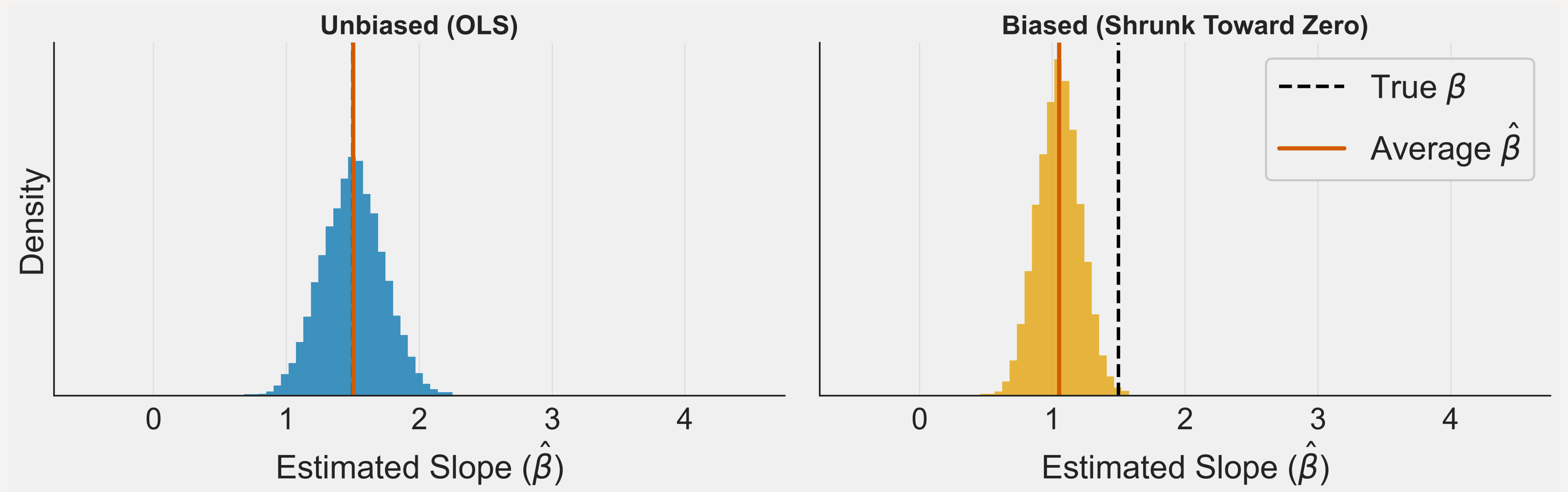
$$Y_i = \frac{\beta_0 X_i}{\beta_1 + X_i} + \varepsilon_i$$

Unbiased

An estimator is unbiased if, averaged over repeated samples, it equals the truth:

$$E \left[\hat{\beta} \right] = \beta$$

Unbiased vs Biased Sampling Distributions



Best

Among all linear unbiased estimators, OLS has the smallest variance in the sampling distribution:

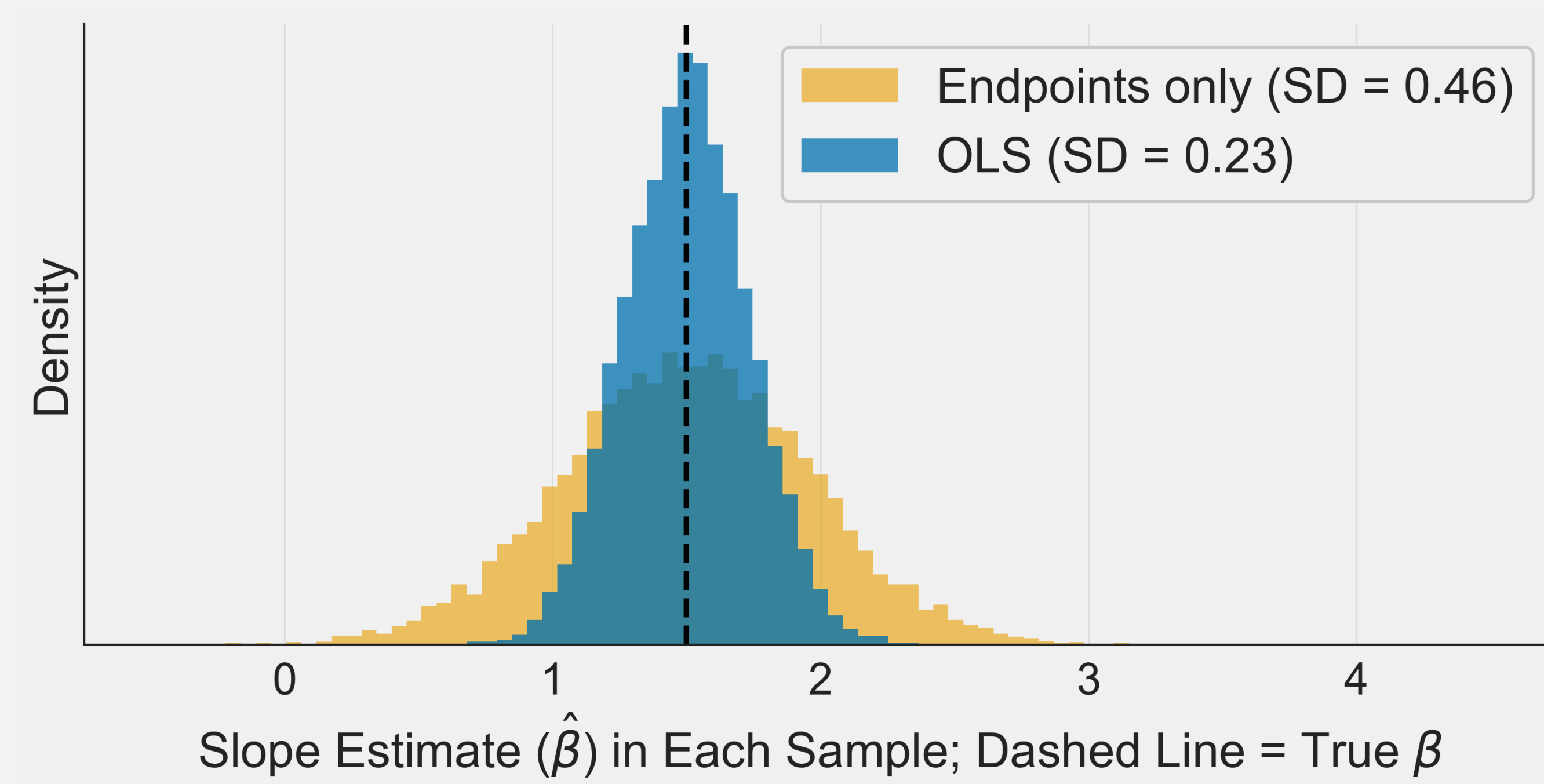
$$\text{Var}(\hat{\boldsymbol{\beta}}_{OLS}) \leq \text{Var}(\tilde{\boldsymbol{\beta}})$$

for any other linear unbiased estimator $\tilde{\boldsymbol{\beta}}$.

Best: The Smallest Variance

Both estimators are linear and unbiased. Each histogram is 10,000 estimates of the same slope.

- **OLS**
- **Endpoints only:** draws a line through the observations with the smallest and largest X and ignores the rest



When Is OLS BLUE?

OLS is BLUE when four assumptions hold.

Gauss-Markov theorem: if the four assumptions hold, no other linear unbiased estimator has a smaller variance.

Today we will cover what the assumptions are, what each one does for us, and what happens when it fails.

Gauss-Markov Assumptions

The Four Assumptions

1. **Linearity** in parameters
2. **Strict exogeneity**
3. **Full rank**
4. **Homoscedasticity**

Assumption 1: Linearity

Linearity in Parameters

The data are generated by the model

$$Y = X\beta + \varepsilon$$

The outcome is a weighted sum of the predictors plus an error.

The weights β are constants.

Linear in the Parameters

Linear in parameters:

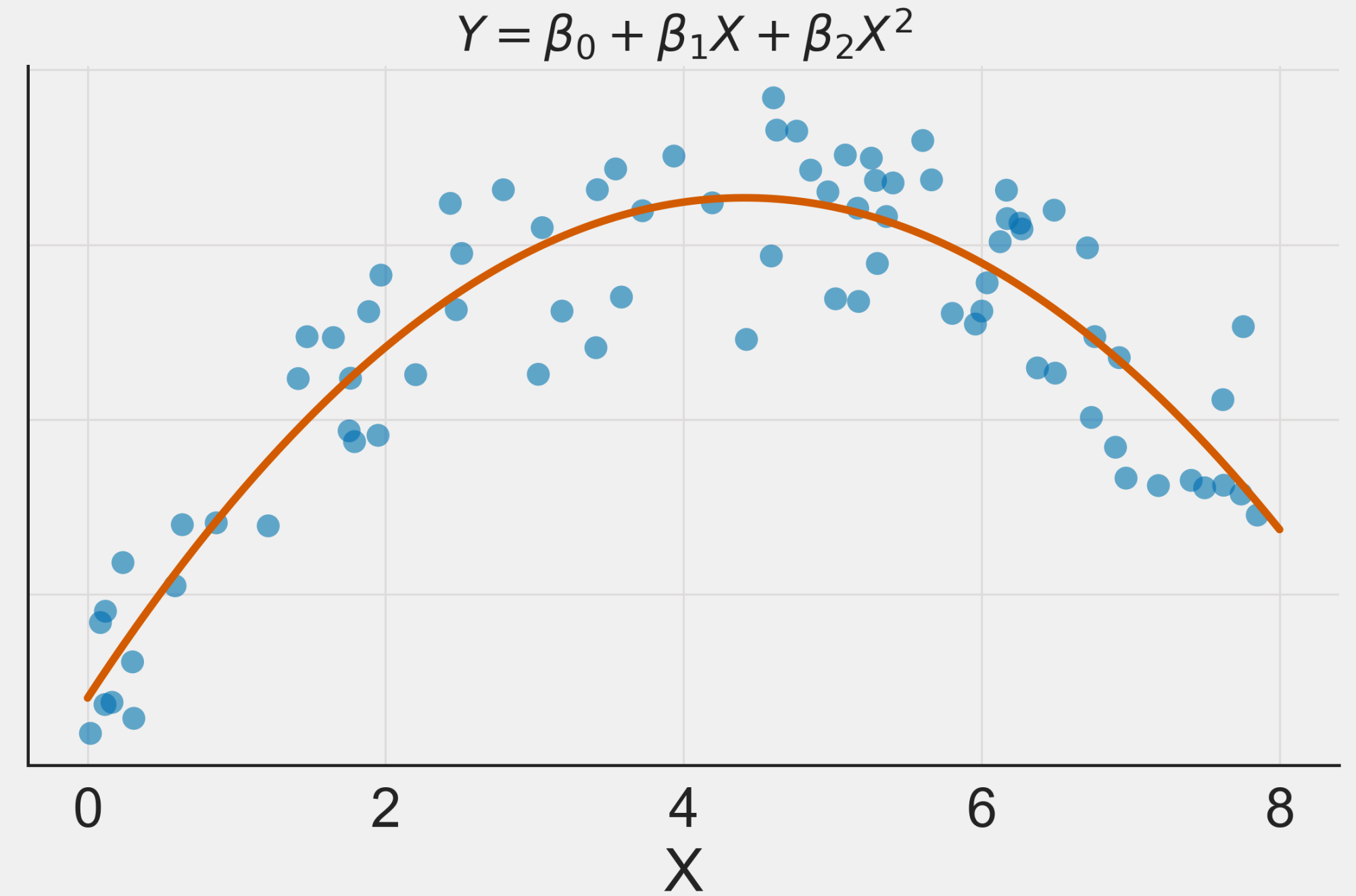
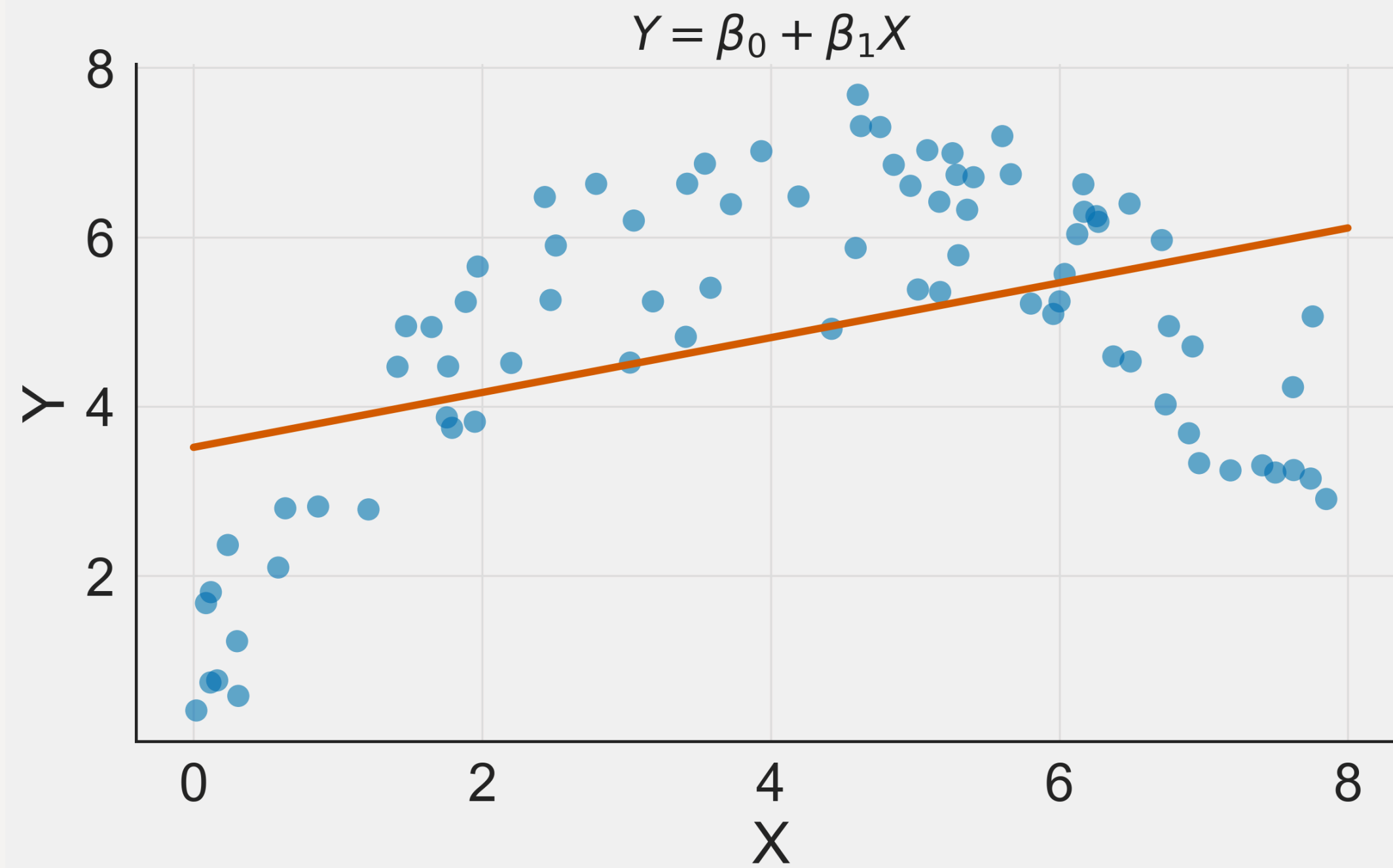
$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i$$

Not linear in parameters:

$$Y_i = \beta_0 + X_i^{\beta_1} + \varepsilon_i$$

Each β must enter as a constant multiplying something we can compute from the data.

A Curve Is Still a Linear Model



Assumption 2: Strict Exogeneity

Strict Exogeneity

$$E [\varepsilon \mid \mathbf{X}] = \mathbf{0}$$

- The errors average to zero at every value of $\mathbf{X}$
- Every predictor is uncorrelated with every error

Other Names

- **Exogeneity**
- **Zero conditional mean:** $E[\varepsilon \mid \mathbf{X}] = \mathbf{0}$
- **Mean independence:** the average error does not depend on $\mathbf{X}$
- **Orthogonality:** the errors are orthogonal to the predictors

Why “Strict” Exogeneity

- **Contemporaneous exogeneity:** error i is unrelated to the predictors of observation i
- **Strict exogeneity:** error i is unrelated to the predictors of every observation
- This is mostly relevant for time series and panel data where errors at time t can affect observations in time $t + 1$.

Errors vs. Residuals

True model: $Y = X\beta + \epsilon$

Estimated model: $Y = X\hat{\beta} + \hat{\epsilon}$

- **Errors** belong to the true model
- **Residuals** belong to the estimated model

Construction vs. Assumption

Residuals are uncorrelated with $\mathbf{X}$ by construction when using OLS. Why?

We set the derivative of SSE to zero to solve for our parameters.

$$\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$$

Errors are uncorrelated with $\mathbf{X}$ by assumption.

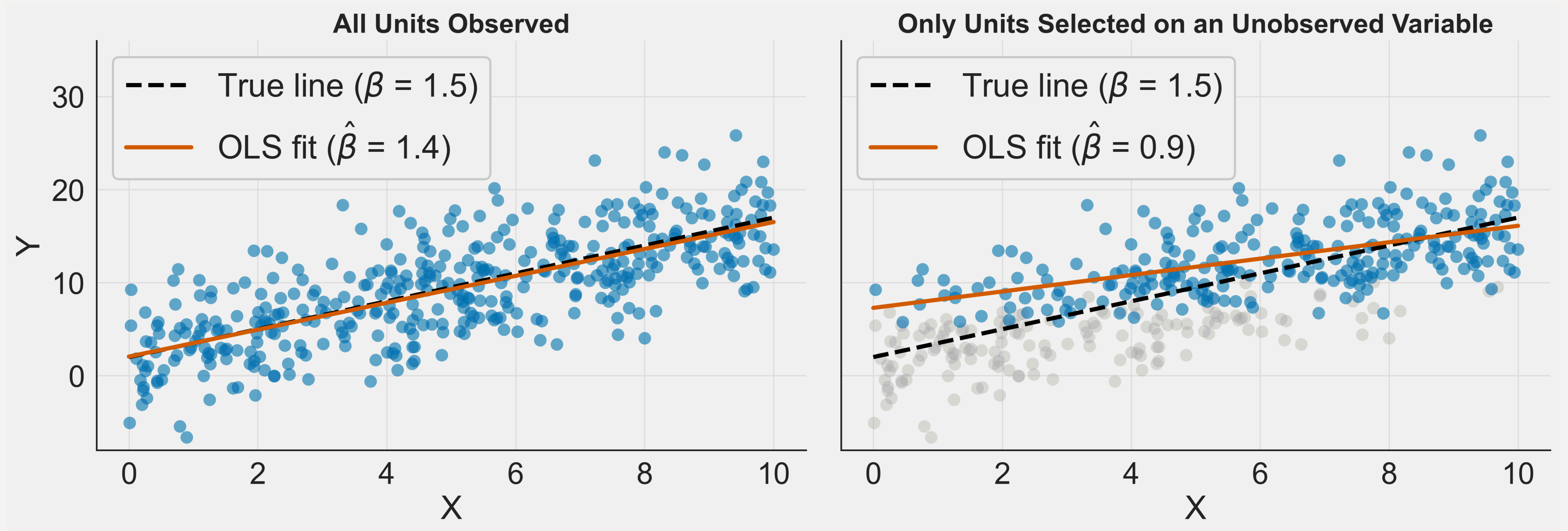
$$E[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$$

The data cannot tell us if this assumption holds.

How Exogeneity Fails

- **Omitted variables:** a confounder affects both X and Y . If omitted, it's absorbed in the error term and ε becomes correlated with X
- **Measurement error:** noise in X is correlated with X
- **Reverse causality:** Y also affects X
- **Wrong functional form:** unmodeled curvature in ε is correlated with X
- **Selection bias:** units enter the sample based on something related to the error, so in the sample the error is correlated with X

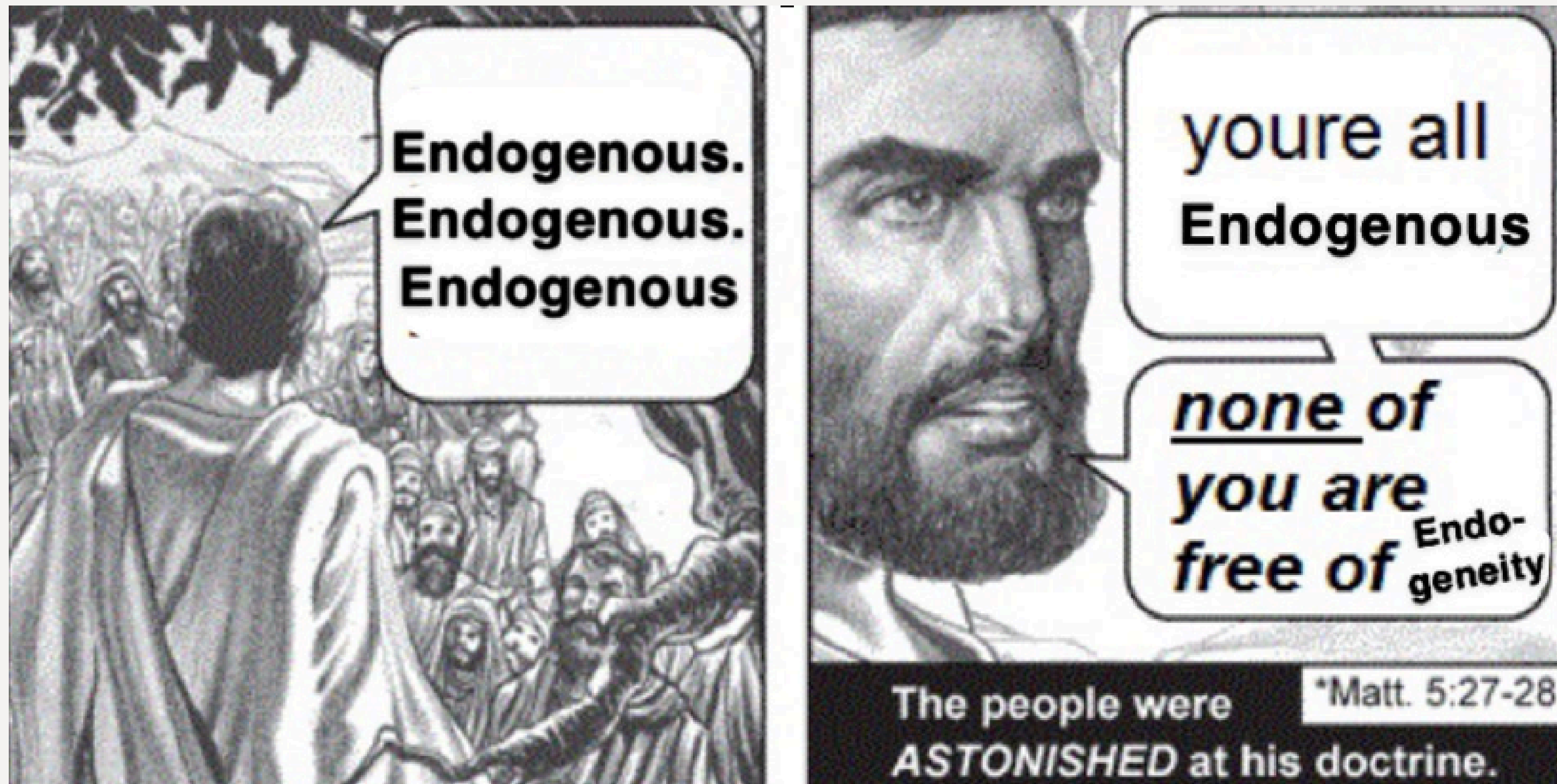
Why It Matters: Bias



Endogeneity

When strict exogeneity fails, X is endogenous.

- **Endogenous:** a predictor that is correlated with the error term
- **Exogenous:** a predictor that is not



Omitted Variable Bias

- $\mathbf{X}_1$: the $n \times k_1$ matrix of included regressors
- $\mathbf{X}_2$: the $n \times k_2$ matrix of omitted regressors

The true model:

$$\mathbf{Y} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}, \quad E[\boldsymbol{\varepsilon} \mid \mathbf{X}_1, \mathbf{X}_2] = \mathbf{0}$$

We only theorize about $\mathbf{X}_1$ so we estimate $\mathbf{Y} = \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 + \hat{\boldsymbol{\varepsilon}}$ with OLS:

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{Y}$$

Step 1: X_2 Moves Into the Error

X_2 does not stop affecting our Y values:

$$Y = X_1 \beta_1 + (X_2 \beta_2 + \epsilon)$$

But since we don't account for it, it moves into the error term of our theorize model. The error term becomes $X_2 \beta_2 + \epsilon$.

Step 2: The Error Is Related to $\mathbf{X}_1$

Strict exogeneity for our theorized model requires $\mathbf{X}_1$ to be orthogonal to the error. Check by looking at the dot product between $\mathbf{X}_1$ and our errors:

$$E \left[\mathbf{X}_1^\top (\mathbf{X}_2 \boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}) \mid \mathbf{X}_1, \mathbf{X}_2 \right] = \mathbf{X}_1^\top \mathbf{X}_2 \boldsymbol{\beta}_2 + \mathbf{X}_1^\top E [\boldsymbol{\varepsilon} \mid \mathbf{X}_1, \mathbf{X}_2]$$

The true error is exogenous, so $E [\boldsymbol{\varepsilon} \mid \mathbf{X}_1, \mathbf{X}_2] = \mathbf{0}$:

$$E \left[\mathbf{X}_1^\top (\mathbf{X}_2 \boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}) \mid \mathbf{X}_1, \mathbf{X}_2 \right] = \mathbf{X}_1^\top \mathbf{X}_2 \boldsymbol{\beta}_2$$

This is not zero unless $\boldsymbol{\beta}_2 = \mathbf{0}$ or $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$. Otherwise strict exogeneity fails.

Step 3: Substitute the True Model

The OLS estimator from the model without $\mathbf{X}_2$:

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{Y}$$

Replace $\mathbf{Y}$ with the true model:

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top (\mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2 + \boldsymbol{\varepsilon})$$

Step 4: Distribute and Simplify

Distribute $(\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top$:

$$\widehat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_1 \boldsymbol{\beta}_1 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 \boldsymbol{\beta}_2 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \boldsymbol{\varepsilon}$$

$(\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_1$ is the identity matrix:

$$\widehat{\boldsymbol{\beta}}_1 = \boldsymbol{\beta}_1 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 \boldsymbol{\beta}_2 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \boldsymbol{\varepsilon}$$

Step 5: Take the Expectation

Given $\mathbf{X}_1$ and $\mathbf{X}_2$, only ε is random. Exogeneity of ε removes the last term:

$$E \left[\widehat{\boldsymbol{\beta}}_1 \mid \mathbf{X}_1, \mathbf{X}_2 \right] = \boldsymbol{\beta}_1 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 \boldsymbol{\beta}_2 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{0}$$

$$\text{Bias} \left(\widehat{\boldsymbol{\beta}}_1 \right) = E \left[\widehat{\boldsymbol{\beta}}_1 \mid \mathbf{X}_1, \mathbf{X}_2 \right] - \boldsymbol{\beta}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 \boldsymbol{\beta}_2$$

Reading the Bias Formula

$$\text{Bias} \left(\widehat{\beta}_1 \right) = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 \beta_2$$

- β_2 : the effect of the omitted regressors on $\mathbf{Y}$
- $(\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2$: the coefficients from regressing each column of $\mathbf{X}_2$ on $\mathbf{X}_1$
- In other words, a portion of $\mathbf{X}_2$ can be explained by $\mathbf{X}_1$. The effect from that portion on $\mathbf{Y}$ is incorrectly attributed to $\mathbf{X}_1$.

No bias if $\beta_2 = \mathbf{0}$ or if $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$.

Direction of the Bias

One included regressor X_1 and one omitted regressor X_2 , with an intercept. The slope on X_1 is biased by

$$\beta_2 \times \frac{s_{1,2}}{s_1^2}$$

	$\text{Cov}(X_1, X_2) > 0$	$\text{Cov}(X_1, X_2) < 0$
$\beta_2 > 0$	Upward	Downward
$\beta_2 < 0$	Downward	Upward

Omitted Variable Bias

1. The omitted regressors go into the error term
2. The error is now correlated with $\mathbf{X}_1$
3. Strict exogeneity fails
4. $\hat{\beta}_1$ is biased

Assumption 3: Full Rank

Full Column Rank

No column of $\mathbf{X}$ is an exact linear combination of the other columns.

Then $\mathbf{X}^\top \mathbf{X}$ is invertible, and

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

exists and is unique.

It also requires more observations than parameters: $n > k + 1$.

What Full Rank Means

Each column of $\mathbf{X}$ should have information that the other columns do not.

Full Rank vs. Not Full Rank

Columns: intercept, X_1 , and a third column.

Full column rank

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 3 & 0 \\ 1 & 4 & 1 \end{bmatrix}$$

Not full rank

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 4 \\ 1 & 3 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

The orange column is exactly 2 times the second column.

Redundant Columns Hide the Effects

With the not-full-rank matrix, the fitted model is

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 (2X_1) = \hat{\beta}_0 + (\hat{\beta}_1 + 2\hat{\beta}_2) X_1$$

Only the combination $\hat{\beta}_1 + 2\hat{\beta}_2$ affects the fit. All of these give the same line:

$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_1 + 2\hat{\beta}_2$
4	0	4
2	1	4
0	2	4

The data can pin down the sum but not the individual coefficients.

Where Rank Enters the Derivation

The normal equations always hold:

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}$$

To solve for $\hat{\boldsymbol{\beta}}$, we multiply both sides by an inverse:

$$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

The orange inverse exists only if $\mathbf{X}$ has full column rank.

Other Names for Full Rank

- No perfect multicollinearity
- Full column rank
- Linearly independent columns
- **Invertibility:** $\mathbf{X}^\top \mathbf{X}$ is invertible (nonsingular)
- **Identification:** β is uniquely determined by the data

Assumption 4: Homoscedasticity

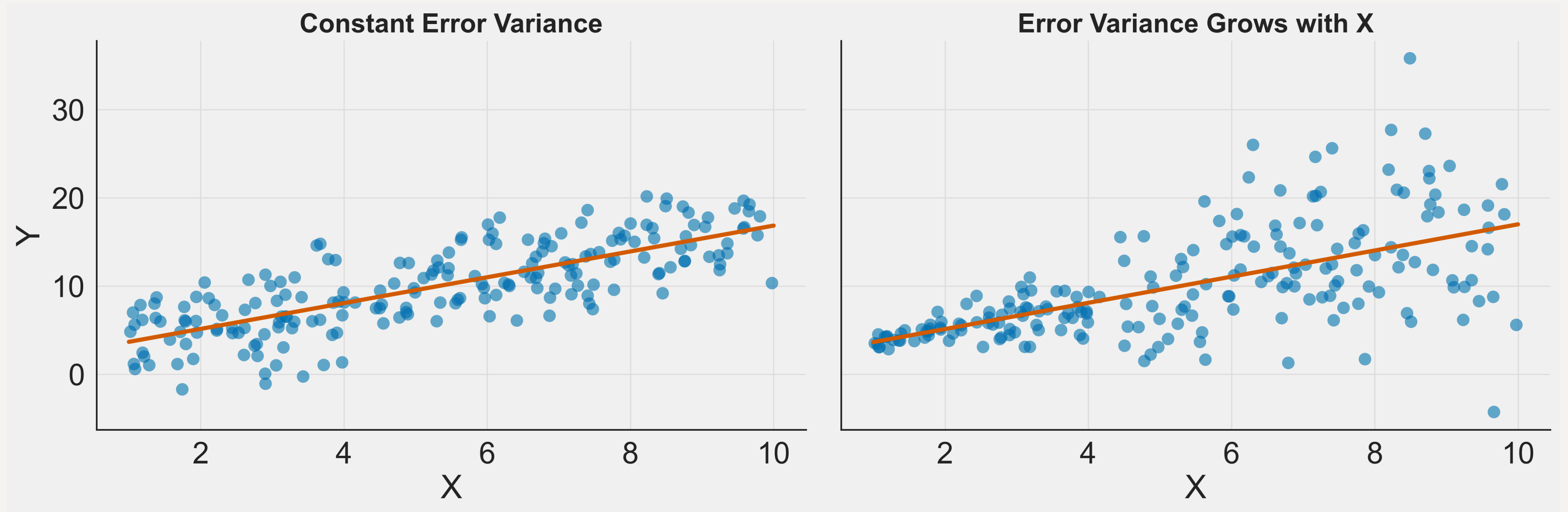
Homoscedasticity

The errors have the same variance at every value of $\mathbf{X}$:

$$\text{Var}(\varepsilon_i \mid \mathbf{X}) = \sigma^2 \quad \text{for every } i$$

When the variance is not constant, the errors are heteroscedastic.

Homoscedastic vs. Heteroscedastic



Other Names for Homoscedasticity

- **Constant error variance**
- **Homogeneity of variance**
- **No heteroscedasticity**
- **Spherical errors:** $\text{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}$
- **No autocorrelation:** the errors are uncorrelated with each other

Common Causes of Heteroskedasticity

- Model Misspecification (omitted variables)
- Outliers
- Nature of the data (e.g. income vs. expenditures)

If Errors Are Heteroscedastic

- OLS is still unbiased
- OLS is no longer best: a weighted estimator can be more precise
- The usual standard errors are wrong, so tests and intervals are off

Fixes: heteroscedasticity-robust standard errors, or weighted least squares.

The Gauss-Markov Theorem

The Theorem

If

1. The model is linear in parameters
2. $E[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$
3. $\mathbf{X}$ has full column rank
4. $\text{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}$

then OLS is BLUE.

Nothing here says the errors are normally distributed.

What Each Assumption Buys Us

Assumption	What it gives us
Linearity	The model we are estimating is the model that generated the data
Strict exogeneity	Unbiasedness
Full rank	A unique solution to the normal equations
Homoscedasticity	The smallest variance, and a simple formula for it

Other Names You Will See

Assumption	Also called
Linearity	Linear in parameters; correct functional form; correct specification
Strict exogeneity	Zero conditional mean; mean independence; exogeneity; orthogonality
Full rank	No perfect multicollinearity; full column rank; invertibility; identification
Homoscedasticity	Constant error variance; spherical errors; no heteroscedasticity

Assumptions for Inference

Three More Assumptions

1. **Validity**
2. **Representativeness**
3. **Normality of errors**

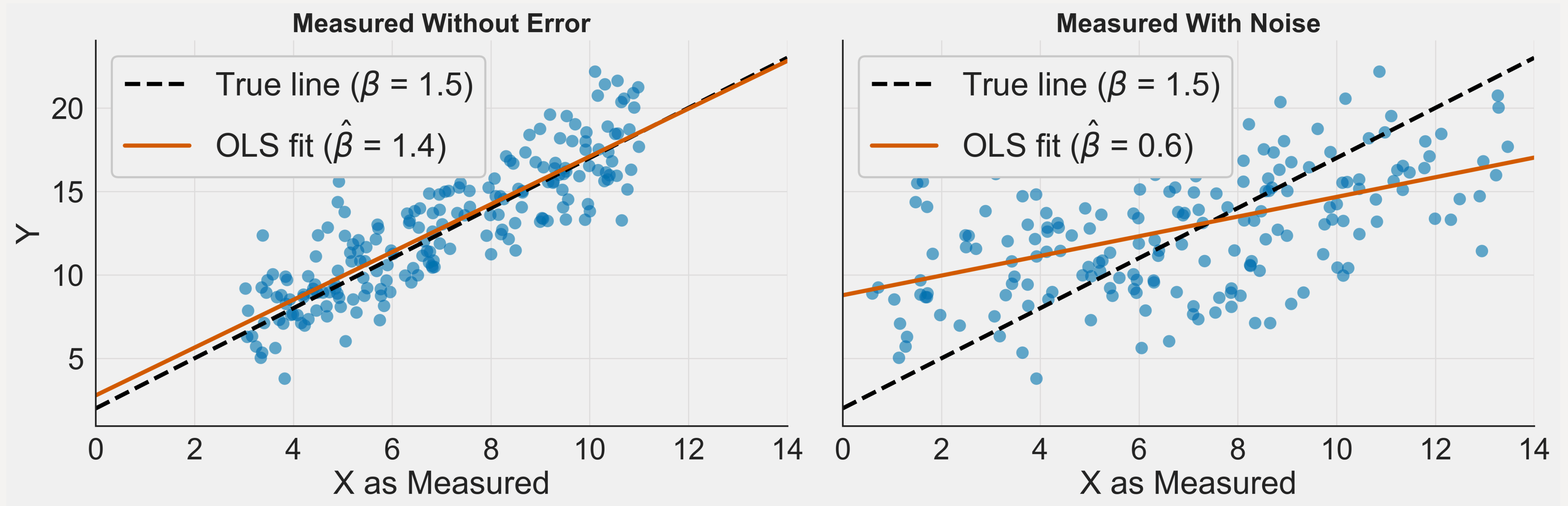
Validity

Validity

The variables in our data measure the concepts in our research question.

- Is party identification a valid measure of ideology?
- Is the number of bills sponsored a valid measure of legislative effectiveness?
- Is self-reported turnout a valid measure of voting?

Invalid Measures Cause Bias



Attenuation bias: Noise in the predictor biases the estimate toward zero

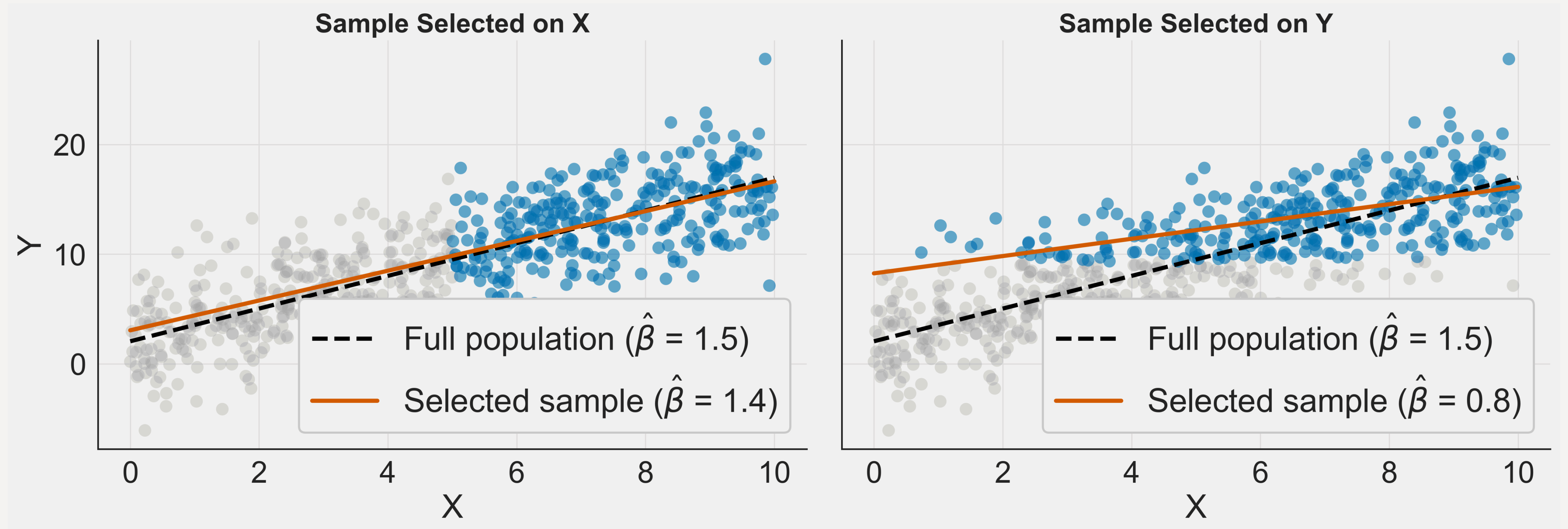
Representativeness

Representativeness

The sample is representative of the population we want to make inferences about.

- Random sampling gives representativeness in expectation
- Nonrandom selection can bias the estimates
- Conclusions apply to the population sampled, not beyond it

Selection on the Outcome



Normality of Errors

Normality of Errors

$$\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$$

Then the sampling distribution of $\hat{\boldsymbol{\beta}}$ is exactly normal:

$$\hat{\boldsymbol{\beta}} \mid \mathbf{X} \sim N(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$$

Where $\hat{\beta}$ Comes From

OLS uses only the data:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

To study how it behaves across samples, assume the data were generated by the model:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

Substitute and simplify:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\beta + \varepsilon) = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon$$

Mean and Variance of $\hat{\beta}$

Given $\mathbf{X}$, the only random part of $\hat{\beta} = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$ is $\boldsymbol{\varepsilon}$, which has mean $\mathbf{0}$ and variance $\sigma^2 \mathbf{I}$.

Mean:

$$E[\hat{\beta} \mid \mathbf{X}] = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top E[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \beta$$

Variance:

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\sigma^2 \mathbf{I}) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$$

Why $\hat{\beta}$ Is Normal

$$\hat{\beta} = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$$

- β is a constant
- $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$ is a fixed weighting of the errors, so each coefficient is a weighted sum of the ε_i

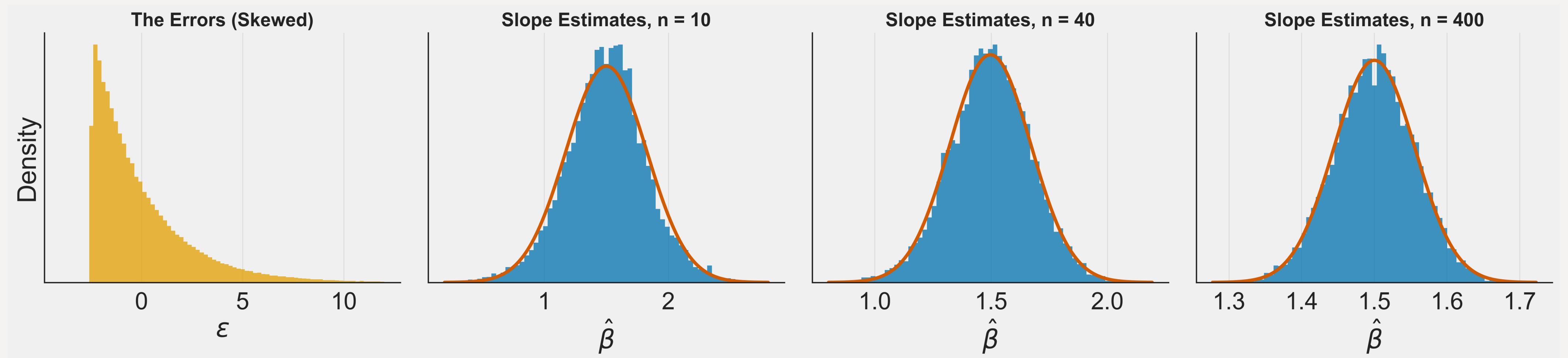
A weighted sum of normal random variables is normal, and adding a constant only shifts the center. So if $\boldsymbol{\varepsilon}$ is normal, $\hat{\beta}$ is normal, with the mean and variance from the last slide.

Normality of Errors

With normal errors:

- t -tests and confidence intervals for individual coefficients are exact
- F -tests for groups of coefficients are exact
- This holds for any sample size

When Errors Are Not Normal



The Central Limit Theorem

Take $W_1, \dots, W_n$ independent, each with mean μ and variance σ^2 , from any distribution. Then as n grows, the sample mean $\overline{W}$ approaches a normal distribution:

$$\overline{W} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

The same holds for weighted sums, as long as no single term dominates. The OLS slope is a weighted sum of the errors, so $\hat{\beta}$ is approximately normal in large samples, even when ε is not.

Summary

- OLS is BLUE: the best (smallest variance) linear unbiased estimator of β
- The Gauss-Markov assumptions are linearity in parameters, strict exogeneity, full rank, and homoscedasticity
- Linearity, strict exogeneity, and full rank give unbiasedness. Homoscedasticity gives the smallest variance
- Strict exogeneity cannot be tested from the data. Failures such as omitted variables produce bias that more data does not fix
- Inference adds validity, representativeness, and normality of errors. Normality matters less as the sample grows

Key Definitions

Key Definitions

- **Estimator**: a rule for computing an estimate from data; it has a sampling distribution
- **BLUE**: best linear unbiased estimator
- **Linear**: linear in the parameters; each β is multiplied by something computed from the data
- **Unbiased**: $E[\hat{\beta}] = \beta$
- **Best**: smallest variance among linear unbiased estimators

Key Definitions

- **Gauss-Markov theorem:** under the four assumptions, OLS is BLUE
- **Linearity in parameters:** $Y = X\beta + \varepsilon$; predictors can be transformed, coefficients enter linearly
- **Strict exogeneity** (zero conditional mean): $E[\varepsilon \mid X] = 0$; errors unrelated to predictors
- **Error vs. residual:** the error is the unobserved deviation in the population; the residual is the computed deviation from the fitted line
- **Endogenous:** a predictor correlated with the error term; exogenous means it is not

Key Definitions

- **Omitted variable bias:** bias from leaving out regressors that affect $\mathbf{Y}$ and are related to the included ones;

$$(\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 \boldsymbol{\beta}_2$$

- **Full rank** (no perfect multicollinearity): no column of $\mathbf{X}$ is an exact linear combination of the others
- **Perfect collinearity:** a violation of full rank; $\mathbf{X}^\top \mathbf{X}$ cannot be inverted

Key Definitions

- **Homoscedasticity** (constant error variance): $\text{Var}(\varepsilon_i \mid \mathbf{X}) = \sigma^2$;
with uncorrelated errors, the errors are spherical
- **Heteroscedasticity**: error variance that changes with $\mathbf{X}$
- **Variance of OLS**: $\text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$

Key Definitions

- **Validity:** the data measure the concept they are meant to measure
- **Attenuation bias:** noise in a predictor biases its coefficient toward zero
- **Representativeness:** the sample reflects the population we want to make inferences about
- **Selection bias:** bias that arises when inclusion in the sample depends on the outcome
- **Normality of errors:** $\varepsilon \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$; gives exact t and F tests in finite samples